The Zeeman Effect

Ben *Morrison*

April 2025

Abstract

The splitting of emission lines in a cadmium lamp were observed when under the influence of an external magnetic field for a net-zero spin transition. Polarisation effects in the emitted light were qualitatively determined for a longitudinal and transversal orientation. To measure the Bohr magneton the magnetic field strength of the external magnetic field was measured as a function of current. By measuring energy splitting as a function of external magnetic field strength a value of the Bohr magneton was determined in both orientations, $\mu_B = 52.2(25)\frac{\mu eV}{T}$ *in the longitudinal orientation and* $\mu_B = 49.7(38)\frac{\mu eV}{T}$ *in the transversal orientation. The average of these two values (mean) is* $\mu_B = 50.95(38)\frac{\mu eV}{T}$.

1 Introduction

In 1896, Pieter Zeeman first observed the splitting of spectral lines lines in a cadmium sample placed under the influence of an external magnetic field, $\underline{B}$ [1]. The magnetic moment of the cadmium atoms interacts with the external magnetic field and the degeneracy of the states is partially lifted. In the normal Zeeman effect, the orientation of the external magnetic field with respect to the direction of the emitted light rays, represented with the wavevector $\underline{k}$, directly affects which degeneracies are lifted, [2]. In this experiment two orientations are considered:

1. longitudinal, $\underline{B} \parallel \underline{k}$
2. transversal, $\underline{B} \perp \underline{k}$

Cadmium has been chosen as the substance to investigate because the dominant state transition $[\mathrm{kr}]_5\mathrm{s}^1\,5\,\mathrm{d}^1 \rightarrow [\mathrm{kr}]_5\mathrm{s}^1\,5\,\mathrm{p}^1$ has a net-zero spin. If this was not the case, the *Anomalous Zeeman effect* would be observed, [3]. It follows directly that the name of the net-zero spin effect is the *Normal Zeeman effect*. The splitting in energy is directly related to the Bohr magneton, see sec. 2.5, and so it can be obtained by a straight line plot in the provided "VideoCom Intensities" software. Thus, the main objectives of the experiment are to observe the Normal Zeeman effect in a cadmium lamp, measure the Bohr magneton and determine the polarisation states of the split emission lines. These objectives will be carried out by using a Fabry-Perot etalon mounted optical train and a VideoCom camera set to "Intensity 2048" mode to measure interference rings on the provided software package, [4].

2 Theory

2.1 Energy Levels in Quantum Mechanics

The wavefunction, $|\psi\rangle$, is a statistical representation of the state of a quantum mechanical particle. It encapsulates all of the information about a quantum mechanical particle. For example, one can find the probability of finding a 1D quantum mechanical particle at a position, x, by computing:

$$\frac{\int_{-\infty}^{+\infty} \langle\psi|x\rangle \langle x|\psi\rangle \, dx}{\langle\psi|\psi\rangle}$$

Or one can find the energy of a particle in a particular state by solving the Schrödinger eigen-equation (time-independent):

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle$$

where $\hat{H}$ is the Hamiltonian operator which acts on a quantum state to return its total energy. It can be shown that the wavefunction for an electron depends on it's distance from the nucleus r, it's azimuthal angle ϕ and it's polar angle θ, [5]:

$$\psi_{n,l,m_l,s,m_s} = \langle r, \theta, \phi | n, l, m_l, s, m_s \rangle$$

which in turn quantises the states with quantum numbers n, the principal quantum number, l, the orbital angular momentum and s, the spin (or intrinsic angular momentum) quantum number. The projection of the orbital and intrinsic angular momenta onto any axis, of which z is the standard, are represented by m_l and m_s. The quantum numbers obey the following rules, [5]:

$$n \in \{1, 2, 3, ..., N\}$$

$$l < |n - 1|$$

$$-|l| \leq m_l \leq |l|$$

The spin quantum number depends on the particle type. For an electron:

$$s = \frac{1}{2}$$

$$m_s = -\frac{1}{2}, \frac{1}{2}$$

When solving the Schrödinger equation for the energy levels of the electrons in the outer shell in the absence of an external magnetic field, $\underline{B}$, one will find that there are many degenerate states (states which share the same energy level). When the cadmium is placed in an external $\underline{B}$ field, the interaction between the electrons magnetic moment, $\underline{\mu}$, and $\underline{B}$ transforms the Hamiltonian into:

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{Zeeman}} \tag{1}$$

where $\hat{H}_0$ is the Hamiltonian neglecting the external magnetic field and the interaction is $\hat{H}_{\text{Zeeman}} = -\underline{\mu}\cdot\underline{B}$. This experiment seeks to observe the splitting in energy levels due to this Zeeman term.

2.2 Polarisation States

An electromagnetic wave's polarisation state refers to a mathematical description of the direction that it's electric field component oscillates in. Two specific types of polarisation are considered in this experiment. An EM wave is linearly polarised if it's electric field oscillates in one direction only, for example:

$$\underline{E} = \underline{e}_1 E_0 e^{i(kz-wt)}$$

oscillates in the $\underline{e}_1$ direction only. A wave is circularly polarised if it can be split into two perpendicular components which have a phase difference of $\frac{\pi}{2}$, for example:

$$\begin{aligned}\underline{E} &= \underline{e}_1 E_0 e^{i(kz-wt-\delta_1)} + \underline{e}_2 E_0 e^{i(kz-wt-\delta_2)} \\ &= \begin{pmatrix} e^{-i\delta_1} \\ e^{-i\delta_2} \end{pmatrix} E_0 e^{i(kz-wt)}\end{aligned}$$

where the total phase difference is $\delta = \delta_1 - \delta_2 = \frac{\pi}{2}$. The column vector above is known as the *Jones vector* and completely describes the polarisation state of an EM wave, [6]. This experiment seeks to qualitatively describe the nature of the polarisation states of the various emissions from the cadmium lamp when subject to an external magnetic field.

2.3 The Normal Zeeman Effect

A special property of the normal Zeeman effect is that the split states are fully polarised (either linearly or circularly). The states are made unqiue by their quantum numbers and the orientation they are in. The states which can be observed in the longitudinal orientation are labelled as follows, [4]:

- σ^+, when $m_l = 1$
- σ^-, when $m_l = -1$

and for the transversal orientation, [4]:

- σ^+, when $m_l = 1$
- σ^-, when $m_l = -1$
- π, when $m_l = 0$

2.4 The Fabry-Perot Etalon

The Fabry-Perot Etalon is an optical device that converts slightly divergent monochromatic light rays into concentric circles. The device is a slab of transparent material which is coated in a reflective coating. The reflective coating is designed specifically so that the wavelength of light which transmits out the back of the etalon is fixed, in this experiment the wavelegnth corresponds to the red Cd line of

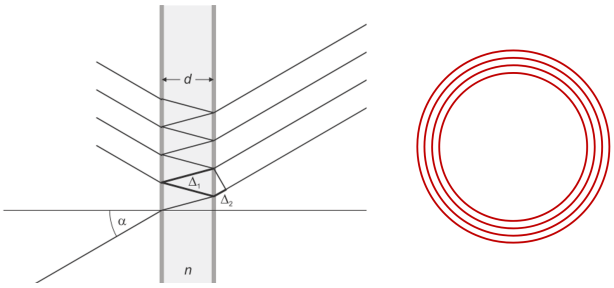


Figure 1: On the left, a diagram of the cross section of a Fabry-Etalon. On the right, an example of the first, second and third interference modes which appear out the back of the etalon when a particular wavelength is focused onto the etalon. The first order mode has the smallest radius, third order has the largest (in the example).

$\lambda = 643.8nm$, [7]. Then the light reflects internally inside the etalon - a small portion of the intensity transmitting back out of the etalon at each reflection. The light which escapes out of the back of the etalon interferes constructively to form a ring pattern out the back of the etalon, as in fig. 1. When the input wavelength changes by a small amount, the aperture of the rings changes - this change can be measured to determine the wavelength of the input light using, [4]:

$$\frac{\Delta\lambda}{\lambda} = \frac{\cos(\beta_2)}{\cos(\beta_1)} - 1 \tag{2}$$

and

$$\frac{\sin(\alpha)}{\cos(\beta)} = n = 1.46 \tag{3}$$

where $\Delta\lambda$ is the shift in wavelength, α_2 is the angle measured from the unperturbed (no external $\underline{B}$) peak and α_1 is the angle of said unperturbed peak. The angle β is a simplification in the notes - by rearranging the equations one can obtain an expression purely in terms of α, which is the directly measured quantity. Also note that the refractive index of the etalon is $n = 1.46$ as stated in [4]. This relationship can then be used to calculate the energy shift by the following relationship, [4]:

$$\Delta E \approx -\frac{\Delta\lambda}{\lambda}\cdot E = -hc\frac{\Delta\lambda}{\lambda^2} \tag{4}$$

where h is Planck's constant and c is the speed of light in vacuum.

2.5 The Bohr Magneton

The Bohr Magneton is a constant which comes from the relationship between an atoms magnetic moment, $\underline{\mu}$, and it's total angular momentum, $\underline{J}$ (from [2]):

$$\underline{\mu} = \frac{e}{2m_e}\underline{J} = \frac{\mu_B}{\hbar}\underline{J} \tag{5}$$

where e is the charge of an electron, m_e is the mass of an electron, $\hbar = \frac{h}{2\pi}$ is the reduced Plancks constant and $\mu_B = \frac{e\hbar}{2m_e}$ is the Bohr magneton. The interaction between $\underline{\mu}$ and an external magnetic field, $\underline{B}$, has energy:

$$E = -\underline{\mu} \cdot \underline{B} = -\frac{eB}{2m_e} J_z = -\frac{eB}{2m_e} M_J \hbar \tag{6}$$

where the subscript z represents a component in the z direction and J and M_J are the total angular momentum quantum numbers from sec. 2.1.

3 The Experiment

3.1 The External Magnetic Field

The external magnetic field was measured using a hall probe in place of the cadmium lamp in the optical train setup. An initial measurement was taken when the magnetic field is off to account for background and then the magnetic field was measured in two ranges. For the longitudinal orientation it was possible to distinguish between the σ^+ and σ^- components for very low currents ($\sim 0.5A$) when compared to the transversal orientation ($\sim 3.1A$) so two different regions were measured. For the longitudinal orientation the magnetic field was measured in increments of $0.5A$ for a range of $0 \rightarrow 5A$. For the transversal orientation the magnetic field was measured in increments of $0.25A$ from a range $3A \rightarrow 5A$. The points measured were used when observing the Zeeman effect so that no interpolation was required.

3.2 Normal Zeeman Effect Observation

To observe the normal Zeeman effect an optical train setup was used, displayed in fig. 2. Before observation could take place the optical setup must be correctly calibrated by adjusting the position of the imaging lens and etalon on the railing until a clear picture could be viewed through the ocular lens. Then the screws on the etalon were adjusted so that the crosshair is roughly at the centre of the concentric rings. After these steps, the ocular lens was switched out for the VideoCom camera and the VideoCom intensities program was setup for use by inputting the focal lens of the condenser lens and centering the pixel positions on the screen by adjusting the zero-point shift parameter (and changing the language from German to English in the VideoCom Intensities program). These calibrations were performed at the start of each lab session as the exact measurements which result in a clearly focused ocular lens picture and centered pixel positions varied each time. The setup could easily be switched between longitudinal and transversal orientations by rotating the lamp (being carful not to touch the metallic base as it is hot) and adjusting the plastic viewing hole. By turning on the magnetic field generator and observing the light intensity peaks on the software, which looked like fig. 4, the splitting of the emission lines from the cadmium lamp were now observable.

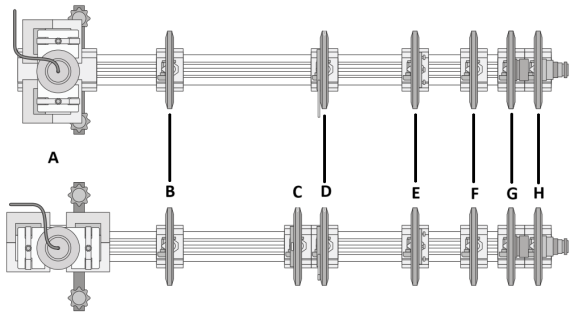


Figure 2: The optical train setup in both configurations as shown from above. The top setup is the transversal orientation and the bottom setup is the longitudinal orientation. A is the cadmium lamp and magnetic field coils; B is the condenser lens (focal length of 150mm); C is a quarter waveplate; D is a linear polariser; E is the Fabry-Perot etalon; F is the imaging lens (focal length of 150mm); G is a colour filter and H is either an ocular lens (for manual observation) or the VideoCom camera (for intensity of light observations via the provided software). Figure is taken from [4] and modified slightly for clarity.

3.3 Determination of Polarisation States

To determine between polarisation states a waveplate and linear polarisers were used.

3.3.1 Longitudinal

In the longitudinal orientation both a waveplate and polariser were used in tandem, 2. By placing a quarter waveplate on the railing, any linearly polarised light would be converted to circularly polarised light and any circularly polarised light would be converted into 45° linearly polarised light, [6]. By placing a linear polariser after the waveplate and checking the $+45^\circ$ and -45° angles, one would expect to see only one of the peaks remain if the input light is circularly polarised. If the input light was linear then both componenets would make it through the linear polariser and two peaks would be expected.

3.3.2 Transversal

In the transversal orientation a linear polariser was placed onto the railling, 2. At 0° it was immediatately observed that the π peak present with no polariser was missing. The polariser was then shifted around to investigate other angles. If all of the components can be filtered by the use of just a linear polariser, this confirms the linear linear nature of σ^+, σ^- and π in the transversal configuration. If a peak is never fully suppressed by the linear polariser then it's polarisaton state is not linear.

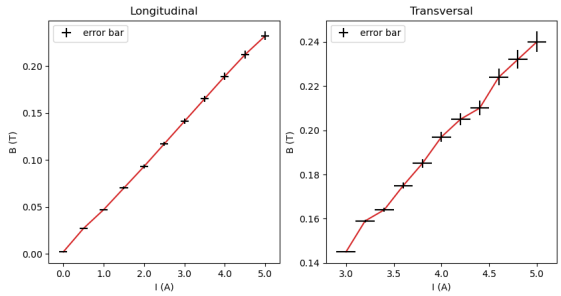


Figure 3: A graph of the magnetic field strength of the external magnet as a function of increasing current. Different regions of current were explored for the longitudinal and transversal orientations. The reason for this is discussed in sec. 3.1. Error bars are included on both the B and I axes in both plots.

3.4 Measurement of the Bohr Magneton

To measure the Bohr magneton the "Zeeman Table" was used in the provided software. It provided a convenient interface which performed all of the calculations discussed in sec. 2.4. The current through the magnetic field generator was incremented regularly in both the longitudinal and transversal configurations ($0A \rightarrow 5A$ for longitudinal and $3A \rightarrow 5A$ for transversal). In the longitudinal orientation the polarisation effects in sec. 3.3 were used so that in the low current regions ($\sim 0A \rightarrow 3.1A$) the σ^+ or σ^- components could be isolated. The same method was not possible for the transversal configuration due to the linear nature of the $\sigma^{\pm}$ components and so the $\sigma^{\pm}$ peaks could not be distinguished until the current in the magnetic field generator was $> 3A$. To record the results, the angle of the π component (always unshifted) and the angle of the shifted $\sigma^{\pm}$ component was recorded and input to the "Zeeman table". The output of this "Zeeman table" along with the raw data are included in the submission, however the pyplot module of the matplotlib python library was used to create the figures in this report.

4 Results

Raw data from the VideoCom Intensities program is submitted in table/graph form directly output from the program itself. Raw data for the external magnetic field is supplied in the form of a spreadsheet. All python scripts used are submitted.

4.1 External Magnetic Field

Whilst not an essential result, the magnetic field strength of the external magnetic field was measured as a function of current, shown in fig. 3.

4.2 Normal Zeeman Effect

The Normal Zeeman Effect was observed qualitatively in fig. 4. In hindsight, it would have been appropriate to take screenshots directly in the VideoCom Intensities program. Alas, I substitute a hand illustration. The circularly polarised components experienced an energy shift under the influence of the external $\underline{B}$ field in both configurations. Linear components were unchanged.

4.3 Bohr Magneton

The literature value of the Bohr Magneton is $\mu_B = 57.9 \frac{\mu eV}{T}$ (form [8]). It will be used for comparison to the values found in this experiment.

4.3.1 Longitudinal Orientation

The Bohr magneton was determined for both the σ^- and σ^+ polarisation states, shown in fig. 5. The clockwise circular polarised component had a gradient of $\mu_B^{\sigma^+} = 49.8(25) \frac{\mu eV}{T}$ and the anti-clockwise circular polarised component had a gradient of $\mu_B^{\sigma^-} = -54.6(25)$. The mean value of these (considering only the modulus), is $\mu_B = 52.2(25) \frac{\mu eV}{T}$. The mean value is within 2.5σ of the literature value.

4.3.2 Transversal Orientation

Similar to the longitudinal orientation, the Bohr magneton was calculated for both of the polarisations separately and then the final value was determined using the mean value, graphically shown in fig. 6. The values in this orientation were $\mu_B^{\sigma^+} = -44.5(38) \frac{\mu eV}{T}$ for the clockwise circular polarised component and $\mu_B^{\sigma^+} = 54.9(38) \frac{\mu eV}{T}$ for the anti-clockwise circular polarised component. The combined value is thus $\mu_B = 49.7(38) \frac{\mu eV}{T}$, which is, again, within 2.5σ of the literature value.

4.3.3 Final Value

Combing the values for the Bohr magneton (by calculating a mean) obtained in the longitudinal and transversal orientations gives a final value of $\mu_B = 50.95(38) \frac{\mu eV}{T}$. The value is within 2σ of the literature value.

4.4 Polarisation states

4.4.1 Longitudinal

In the absence of any polarisers the π component was never observed. A quarter waveplate was placed in front of a linear polariser on the optical train. When the waveplate was set to $0°$ and the linear polariser set to $0°$, the σ^+ and σ^- components were both detected by the VideoCom Camera. By rotating the linear polariser to $+45°$ one could observe only the σ^+ component. Similarly, by rotating the linear

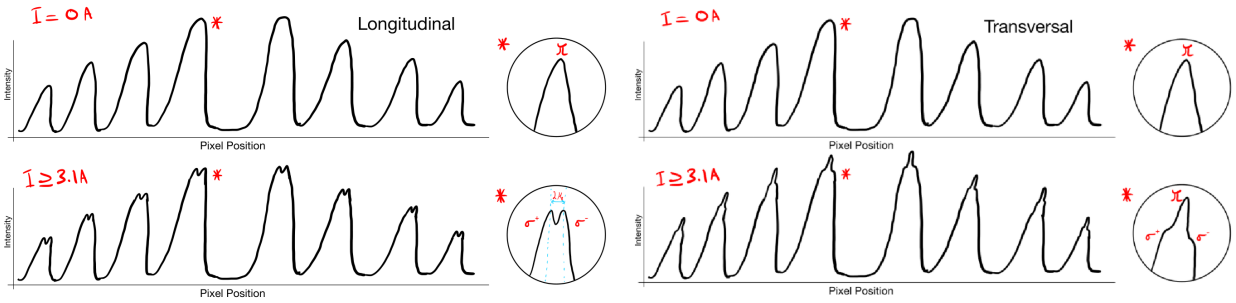


Figure 4: An example of the reading shown in the VideoCom Intensities program in both orientations before and after applying an external magnetic field. Without the use of polarisers in the optical train setup, no visible change to the peaks was observed until a current of $I \geq 3.1A$ was used to drive the external magnetic field. On the right sides of the figures the intensity peaks corresponding to the σ^+, σ^- and π polarisation states have been labelled.

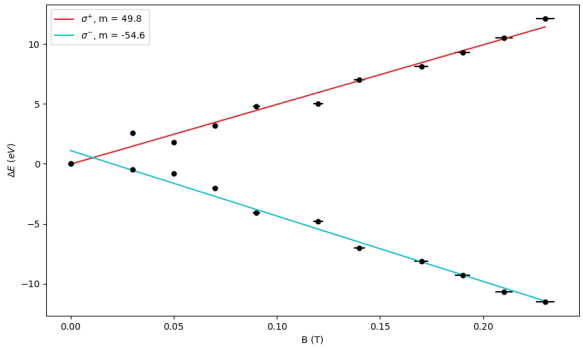


Figure 5: A graph of external magnetic field strength against the observed energy shift of the cadmium lamp emission with the setup in the longitudinal orientation. The red line represents clockwise circular polarisation and the blue line represents anti-clockwise circular polarisation of the cadmium emission. Errors bars are included in both the B and ΔE axes.

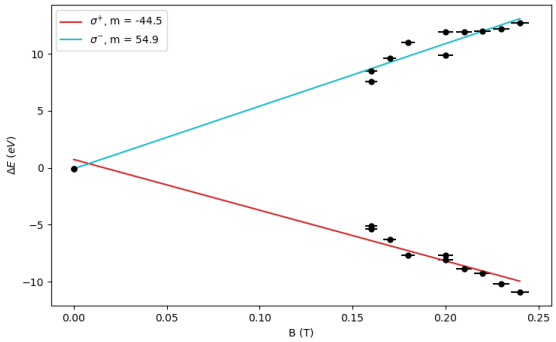


Figure 6: A graph of external magnetic field strength against the observed energy shift of the cadmium lamp emission with the setup in the transversal orientation. The red line represents clockwise circular polarisation and the blue line represents anti-clockwise circular polarisation of the cadmium emission. Errors bars are included in both the B and ΔE axes.

polariser to $-45°$ only the σ^- component could be observed. The quarter waveplate converts the assumed to be circularly polarised light into linearly polarised light and then the linear polariser blocks out one handedness or the other. Thus, directly confirming the circular polarisation of the σ^+ and σ^- components in the longitudinal orientation.

4.5 Transversal

In the absence of any polarisers one could observe the σ^+, σ^- and the π components. By inserting a a linear polariser at $0°$, the π component was no longer visible. Setting the polariser to $90°$ resulted in the σ^+ and σ^- components no longer being detected and the π component reappearing. This confirms the linear polarisation state of all three components in the transversal orientation.

5 Discussion

5.1 Sources of Experimental Error

5.1.1 Hall Probe

The manual for the Hall Probe, [9], states that the resolution of the device is $0.01mT$ or $0.1mT$. The $0.01mT$ mode was used for measurements of the external magnetic field strength. For absolute measurements the manual quotes the accuracy of the hall probe to be $\pm 2\%$ and for relative measurements $\pm 0.5\%$. In this case the experiment is concerned with the absolute value of the magnetic field strength at the position of the lamp so an uncertainty of $\pm 2\%$ is chosen for magnetic field measurements. This means that the error scales linearly with the magnetic field strength. This effect is visible in fig. 3, where the uncertainties are considerably larger towards the higher end of the B readings. It seems from the manual that there is little that can be done about this for absolute magnetic field strength measurements due to a lack of a standard reference field to calibrate the probe.

5.2 Angle Measurements

Angle measurements were performed using the VideoCom Intensities software. No recordings were taken for the uncertainties provided by the software for each measurements as it was assumed this would be recorded in the table that the program outputs. Alas, after consulting a colleague I was informed that the average uncertainty was just under $\sim 0.05°$. This limitation likely comes from the physical dimensions of the pixels and their separation, [10], but no greater detail is provided in the manuals on this source of error. Background fluctuations are reported in the videoCom manual due to constructive and destructive interference at the lens of the camera and a light-sensitive area inside the camera. These fluctuations affect the absolute light intensity measurement, which this experiment is not concerned with, and so this can be neglected. When the setup was in the transversal orientation there was a larger uncertainty associated with low B readings because the (Gaussian) peaks corresponding to σ^+ and σ^- components are very much overlapping. This behaviour is present over the entire range of measurements but lessens as B increases. In the longitudinal orientation this error was entirely suppressed because the polarisation effects discussed in sec. 3.3 ensure that only σ^+ or σ^- are incident on the camera (depending on the orientation of the polarisers). With no way to quantify these contributions a flat value of $\sigma_\alpha = 0.05$ was chosen and used in the error propagation steps for other variables.

5.3 Optical Alignment

When setting up the optical train, adjusting the imaging lens and etalon to get a clear picture proved difficult. The experimental setup was also used by other students throughout the week. Despite efforts being made to ensure the optical alignment was similar throughout sessions (i.e. noting positions of components on the rails), the interference rings were blurry at best when viewed through the optical lens. No amount of imaging lens, condensing lens or etalon adjustment could fix this. This effect likely broadened the (Gaussian) intensity peaks and should have contributed through the uncertainty provided by the VideoCom Intensity program.

5.4 Energy Shift

The energy shift is calculated using eqn. 4. By applying the standard calculus error propagation formula:

$$\sigma_{\Delta E} = \sqrt{(\delta\alpha_1)^2 \left(\frac{\partial \Delta E}{\partial \alpha_1}\right)^2 + (\delta\alpha_2)^2 \left(\frac{\partial \Delta E}{\partial \alpha_2}\right)^2} \quad (7)$$

the formula for the error in energy shift is obtained (working in lab book). A python script is then used to compute all of the values which are outputted as a python list to be used in plotting and further error propagation calculations.

5.5 Bohr Magneton

The Bohr Magneton was determined using a mean value between two gradient measurements from fig. 5 and fig. 6. A covariance matrix was calculated using the Numpy library's poly1D and polyfit functions. From this the standard deviation was obtained by taking the square root of the diagonal components. These values are the final quoted uncertainties in the results section. Only one good set of measurements (measurements which did not have an experimental setup issue) was obtained before the conclusion of the third (and final) laboratory session. If it was possible to attend the fourth laboratory slot, measurements could have been refined and more data collected to reduce these errors quite significantly. For example, sampling the magnetic field at finer intervals, performing repeat measurements and noting down the reported uncertainties in angle measurements for each measurement would all contribute to lowering the final uncertainties.

5.6 Residuals

The residuals, shown in fig. 7, show that a lot more than $\frac{2}{3}$ of the points are within one σ. This means that the errors are significantly underestimated. Also present is a correlation with B. This residuals plot is not totally unexpected considering the error on angle measurements is a crude estimate (and the calculation is completely ignorant of the effect of optical misalignment), which has a compound effect on the error in energy shift, ΔE, and thus the Bohr magneton value.

6 Conclusion

In conclusion, the Zeeman effect was observed in a cadmium lamp when placed under the influence of an external magnetic field. The dominant transition had a net-zero spin so the Normal Zeeman effect was observed. A special property of the Normal Zeeman effect is that the emitted photons are polarised. Using an optical train setup with a waveplate and linear polariser, the polarisation states were qualitatively determined. The magnetic field strength at the position of the lamp was measured as a function of current through the coils. This information was used to plot a graph of energy shift against magnetic field strength. The gradient of this graph is a constant value called the Bohr magneton. The value for the Bohr magneton was obtained to within 2σ of the literature value, from [8]. The uncertainty estimations suggest that the data obtained is very unreliable, with the most significant contribution being a lack of data collection from the VideoCom Intensity software. Reliability of the results could be improved in the event of the experiment being repeated by: noting down the uncertainty values for each measurement of angle in the VideoCom Intensity software

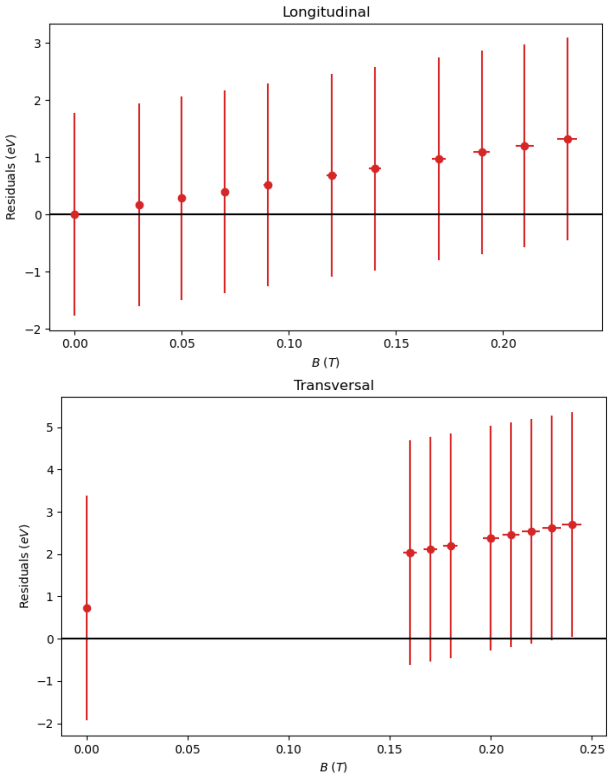


Figure 7: A graph of the residuals for the energy shift values in both the transversal and longitudinal orientations. The x-axis is the magnetic field strength measured in Teslas (T).

and taking finer measurements of the magnetic field strength as a function of current.

References

[1] Jeremy Tatum. *Zeeman effect*. URL: https://phys.libretexts.org/Bookshelves/Astronomy__Cosmology/Stellar_Atmospheres_(Tatum)/07%3A_Atomic_Spectroscopy/7.21%3A_Zeeman_effect. (24/3/2025)

[2] School of Physics Astronomy. "The Zeeman Effect". *Junior Honours Lab Scripts*. (2024)

[3] Thomas Preston. "Radiation phenomena in a strong magnetic field". *The Scientific Transactions of the Royal Dublin Society*. (1898)

[4] LEYBOLD Physics Leaflets. "Leybold-Didactic GmbH Leaflet P6.2.7.3". *Observing the normal Zeeman effect in transverse and longitudinal configuration*

. [5] Nouredine Zettili. "Quantum Mechanics: Concepts and Applications". *Quantum Mechanics: Concepts and Applications, by Nouredine Zettili, pp. 664. ISBN 0-471-48944-1. Wiley-VCH , October 2001.* (2001)

[6] Eugene Hecht. *Optics Global Edition*. P. 728. (2016)

[7] LEYBOLD Physics Leaflets. "Instruction sheet 471 221". *Observing the normal Zeeman effect in transverse and longitudinal configuration*

. [8] LEYBOLD Physics Leaflets. "Leybold-Didactic GmbH Leaflet P6.2.7.4". *Measuring the Zeeman splitting of the red cadmium line as a function of the magnetic field –spectoscopy using a Fabry-Perot etalon*

. [9] LEYBOLD Physics Leaflets. "Instruction manual for EP-15 Flux Density Meter". *Daedalon Corporation*

. [10] LEYBOLD Physics Leaflets. "VideoCom Retroreflecting Foil Falling Body for VideoCom". *LEYBOLD DIDACTIC GMBH*